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Clostridium acetobutylicum

Clostridium acetobutylicum, ATCC 824, is a commercially valuable bacterium sometimes called the "**Weizmann Organism**", after Jewish Russian-born biochemist Chaim Weizmann. A senior lecturer at the University of Manchester, England, he used them in 1916 as a bio-chemical tool to produce at the same time, jointly, acetone, ethanol, and n-butanol from starch. The method has been described since as the ABE process, (Acetone Butanol Ethanol fermentation process), yielding 3 parts of acetone, 6 of *n*-butanol, and 1 of ethanol. Acetone was used in the important wartime task of casting cordite. The alcohols were used to produce vehicle fuels and synthetic rubber.

Unlike yeast, which can digest only some sugars into alcohol and carbon dioxide, *C. acetobutylicum* and other Clostridia can digest whey, sugar, starch, cellulose and perhaps certain types of lignin, yielding *n*-butanol, propionic acid, ether, and glycerin.

In genetic engineering

In 2008, a strain of *Escherichia coli* was genetically engineered to synthesize butanol; the genes were derived from *Clostridium acetobutylicum*.^{[1][2]} In 2013, the first microbial production of short-chain alkanes was reported^[3] - which is a considerable step toward the production of gasoline. One of the crucial enzymes - a fatty acyl-CoA reductase - came from *Clostridium acetobutylicum*.

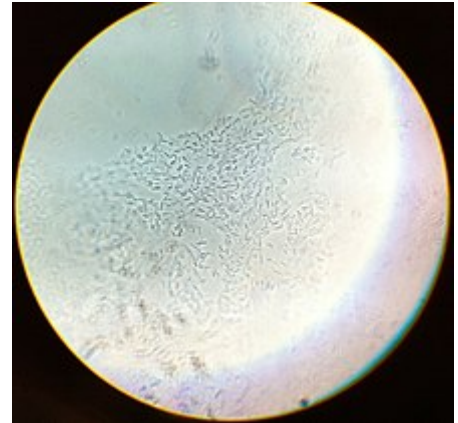
See also

- ABE
- Acetone
- Butanol
- Clostridium beijerinckii
- Ethanol

References

- M. Goho, Alexandra (2008-01-16). "Better Bugs for Making Butanol" (<http://www.technologyreview.com/news/409400/better-bugs-for-making-butanol/>). *MIT Technology Review*.

Clostridium acetobutylicum



Scientific classification

Domain:	<u>Bacteria</u>
Phylum:	<u>Bacillota</u>
Class:	<u>Clostridia</u>
Order:	<u>Eubacteriales</u>
Family:	<u>Clostridiaceae</u>
Genus:	<u><i>Clostridium</i></u>
Species:	<i>C. acetobutylicum</i>

Binomial name

Clostridium acetobutylicum
McCoy *et al.* 1926 (Approved Lists
1980)



Chaim Weizmann

2. Atsumi, S.; Hanai, T.; Liao, JC. (Jan 2008). "Non-fermentative pathways for synthesis of branched-chain higher alcohols as biofuels". *Nature*. **451** (7174): 86–9. Bibcode:2008Natur.451...86A (<https://ui.adsabs.harvard.edu/abs/2008Natur.451...86A>). doi:10.1038/nature06450 (<https://doi.org/10.1038%2Fnature06450>). PMID 18172501 (<https://pubmed.ncbi.nlm.nih.gov/18172501>). S2CID 4413113 (<https://api.semanticscholar.org/CorpusID:4413113>).
3. Choi, YJ.; Lee, SY. (Oct 2013). "Microbial production of short-chain alkanes". *Nature*. **502** (7472): 571–4. Bibcode:2013Natur.502..571C (<https://ui.adsabs.harvard.edu/abs/2013Natur.502..571C>). doi:10.1038/nature12536 (<https://doi.org/10.1038%2Fnature12536>). PMID 24077097 (<https://pubmed.ncbi.nlm.nih.gov/24077097>). S2CID 4393929 (<https://api.semanticscholar.org/CorpusID:4393929>).

Further reading

- Nölling J, Breton G, Omelchenko MV, et al. (August 2001). "Genome sequence and comparative analysis of the solvent-producing bacterium *Clostridium acetobutylicum*" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC99537>). *J. Bacteriol.* **183** (16): 4823–38. doi:10.1128/JB.183.16.4823-4838.2001 (<https://doi.org/10.1128%2FJB.183.16.4823-4838.2001>). PMC 99537 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC99537>). PMID 11466286 (<https://pubmed.ncbi.nlm.nih.gov/11466286>).
- Driessen AJ, Ubbink-Kok T, Konings WN (February 1988). "Amino acid transport by membrane vesicles of an obligate anaerobic bacterium, *Clostridium acetobutylicum*" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC210727>). *J. Bacteriol.* **170** (2): 817–20. doi:10.1128/jb.170.2.817-820.1988 (<https://doi.org/10.1128%2Fjb.170.2.817-820.1988>). PMC 210727 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC210727>). PMID 2828326 (<https://pubmed.ncbi.nlm.nih.gov/2828326>).
- Zappe H, Jones WA, Jones DT, Woods DR (May 1988). "Structure of an endo-beta-1,4-glucanase gene from *Clostridium acetobutylicum* P262 showing homology with endoglucanase genes from *Bacillus* spp" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC202643>). *Appl. Environ. Microbiol.* **54** (5): 1289–92. Bibcode:1988ApEnM..54.1289Z (<https://ui.adsabs.harvard.edu/abs/1988ApEnM..54.1289Z>). doi:10.1128/AEM.54.5.1289-1292.1988 (<https://doi.org/10.1128%2FAEM.54.5.1289-1292.1988>). PMC 202643 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC202643>). PMID 3389820 (<https://pubmed.ncbi.nlm.nih.gov/3389820>).
- Bowles LK, Ellefson WL (November 1985). "Effects of butanol on *Clostridium acetobutylicum*" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC238718>). *Appl. Environ. Microbiol.* **50** (5): 1165–70. Bibcode:1985ApEnM..50.1165B (<https://ui.adsabs.harvard.edu/abs/1985ApEnM..50.1165B>). doi:10.1128/AEM.50.5.1165-1170.1985 (<https://doi.org/10.1128%2FAEM.50.5.1165-1170.1985>). PMC 238718 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC238718>). PMID 2868690 (<https://pubmed.ncbi.nlm.nih.gov/2868690>).
- US 1875536 (<https://worldwide.espacenet.com/textdoc?DB=EPODOC&IDX=US1875536>)
- US 1315585 (<https://worldwide.espacenet.com/textdoc?DB=EPODOC&IDX=US1315585>)
- Weber, Christian; Farwick, Alexander; Benisch, Feline; Brat, Dawid; Dietz, Heiko; Subtil, Thorsten; Boles, Eckhard (10 June 2010). "Trends and challenges in the microbial production of lignocellulosic bioalcohol fuels". *Applied Microbiology and Biotechnology*. **87** (4): 1303–1315. doi:10.1007/s00253-010-2707-z (<https://doi.org/10.1007%2Fs00253-010-2707-z>). ISSN 0175-7598 (<https://www.worldcat.org/issn/0175-7598>). PMID 20535464 (<https://pubmed.ncbi.nlm.nih.gov/20535464>). S2CID 12601202 (<https://api.semanticscholar.org/CorpusID:12601202>).
- Jones, DT; Woods, DR (1986). "Acetone-butanol fermentation revisited" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC373084>). *Microbiological Reviews*. **50** (4): 484–524. doi:10.1128/MMBR.50.4.484-524.1986 (<https://doi.org/10.1128%2FMMBR.50.4.484-524.1986>)

86). PMC 373084 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC373084>). PMID 3540574 (<https://pubmed.ncbi.nlm.nih.gov/3540574>).

- Bartha, Ronald M. Atlas & Richard (1993). *Microbial ecology : fundamentals and applications* (<https://archive.org/details/microbialecology00atla/page/563>) (3rd ed.). Redwood City, Calif.: Benjamin/Cummings Pub. Co. p. 563 (<https://archive.org/details/microbialecology00atla/page/563>). ISBN 978-0-8053-0653-8.
- *Microbial Processes: Promising Technologies for Developing Countries* (http://www.nap.edu/openbook.php?record_id=9544&page=R1). Washington: National Academy of Sciences. 1979. doi:10.17226/9544 (<https://doi.org/10.17226%2F9544>). ISBN 978-0-309-57050-3. Retrieved 10 May 2011.
- Wong Kromhout, Wileen (2011-03-16). "UCLA researchers engineer E. coli to produce record-setting amounts of alternative fuel" (<https://archive.today/20150418144123/http://newsroom.ucla.edu/releases/ucla-engineers-drive-e-coli-to-197675>). *UCLA Newsroom*. Archived from the original (<http://newsroom.ucla.edu/releases/ucla-engineers-drive-e-coli-to-197675>) on 2015-04-18. Retrieved 2015-04-18.

External links

- ATCC reference organism 824 C.Acetobutylicum. (<http://www.atcc.org/ATCCAdvancedCatalogSearch/ProductDetails/tabid/452/Default.aspx?ATCCNum=824&Template=bacteria>) Archived (<https://web.archive.org/web/20120318094452/http://www.atcc.org/ATCCAdvancedCatalogSearch/ProductDetails/tabid/452/Default.aspx?ATCCNum=824&Template=bacteria>) 2012-03-18 at the Wayback Machine
- findarticles.com: Bacteria speeds drug to tumors - use of Clostridium acetobutylicum enzyme to activate cancer drug CB 1954 (http://www.findarticles.com/p/articles/mi_m1272/is_n2593/v123/ai_15826295).
- EPA Clostridium acetobutylicum Final Risk Assessment (http://www.epa.gov/biotech_rule/pubs/fra/fra003.htm)
- Genetic Engineering of Clostridium acetobutylicum for Enhanced Production of Hydrogen Gas (https://web.archive.org/web/20060528013505/http://www.engr.psu.edu/h2e/Pub/Regan/Regan_etal_1.htm): Penn State University.
- Pathema-Clostridium Resource (<https://archive.today/20121204183216/http://pathema.tigr.org/tigr-scripts/Clostridium/PathemaHomePage.cgi>)
- Chaim Weizmann (http://www.encyclopedia.com/topic/Chaim_Weizmann.aspx)
- Type strain of *Clostridium acetobutylicum* at BacDive - the Bacterial Diversity Metadatabase (<http://bacdive.dsmz.de/index.php?search=2529&submit=Search>)

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